Mohammed Alkindi

Computer Science | Data Analytics | Software Engineering | Open Source
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EXPERIENCE

Open Source Contributor Aug 2026 - Present

- Merged **169 pull requests across 43+ external repositories** (39+ at Microsoft), including projects maintained by NVIDIA, AWS, Google, Hugging Face, OpenAI, and Pydantic. Most active in simple-statistics, Microsoft hve-core, and NegPy.
- Fixed floating-point cancellation in simple-statistics linear regression that returned a slope 69% wrong on timestamp inputs, by centering sums on the means. Applied the same fix to Jenks variance.
- Fixed a Microsoft onnxscript optimizer rule that ignored Gemm's beta when folding BatchNorm into Gemm, so rewritten ONNX models silently returned wrong values.
- Fixed PyTorch distributed-training bugs in NVIDIA NeMo, including unreduced validation loss across ranks.

Shell | Data Analyst Intern Jun 2026 - Aug 2026

- Automated the daily production report with a Python pipeline (pandas, openpyxl, matplotlib) that auto-detects 2 report layouts and renders the dashboard in under 5 seconds, replacing roughly 3 hours of manual Excel work per day.
- Wrote 12 fail-closed validation rules (types, ranges, duplicates, required metrics) that block publishing on bad input, and logged a SHA-256 hash of every source file for audit.
- Built an interactive dashboard with asset filters, well search, and 4 KPI cards, plus PDF and email outputs.

PROJECTS

ProofX | Python, Numba, Lean 4 proofx.org

- Built a parallel counterexample search engine in Python and Numba that checks ~21M integers per second on 2 CPU cores, about 200x faster than plain Python, with every run logged to a seeded ledger so results replay exactly.
- Converted search ledgers into 500 Lean 4 theorems verified by the Lean kernel, and automated conjecture testing by racing 7 proof tactics and 3 counterexample searches under a 15-second deadline. Covered by 437 passing tests.

Ridge | Node.js, Express, Claude API, Vercel tryridge.vercel.app

- Spreadsheet analysis app that computes 10+ statistical methods (Pearson, Spearman, Welch, chi-square, KS) in code before any LLM sees the data, returning results in under 100 ms on typical files.
- Parses 11 file formats and records method, sample size, and source rows for every finding. Covered by 565 unit and API tests plus Playwright browser tests, with security scanning and rate limiting.

Switchyard | TypeScript, Node.js, npm @switchyardhq/switchyard

- CLI with 19 commands that runs multiple AI coding agents in isolated Git worktrees and flags merge conflicts early by simulating every agent pair's merge in memory with git merge-tree.
- Shipped 4 npm releases with 260 integration tests against real Git repos, verified in CI across Windows, macOS, and Linux on 4 Node versions.

EDUCATION

University of California, San Diego | B.S. Computer Science

University of Pennsylvania | Professional Development Program

HONORS AND LEADERSHIP

- **Thiel Fellowship Semifinalist (2026):** advanced to the semifinal round of the $250K fellowship for founders building outside school, top ~3% of 3,000+ global applicants.
- **IDEA Scholar, UC San Diego Jacobs School of Engineering (2026):** competitive, limited-seat multi-year cohort.
- **Oman Government Scholarship (2025):** fully funded award from the Ministry of Higher Education covering tuition and a monthly stipend, valued at ~$400K over five years.
- **Mathematics Student of the Year (2025):** sole recipient, and top 4 team nationally in mathematics (2024).
- **Meta Connect 2026:** invited by Meta to attend in Menlo Park and join its AI Developer Roundtable.
- **VP of Outreach, Penn Chess Club (2025 - 2026):** lifted engagement and retention 40%+ and built the club website.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, Bash, PowerShell, Lean 4
Data and ML: pandas, NumPy, scikit-learn, PyTorch, XGBoost, statistical inference, K-Means, PCA
Frameworks and Tools: Node.js, Express, FastAPI, React, PostgreSQL, Supabase, Docker, Git, GitHub Actions, pytest, Playwright
Certifications: Machine Learning Specialization (Stanford), CS50x and CS50P (Harvard) | **Spoken:** Arabic, English (native)